UNIVERSITY EXAMINATION

INF221 – Web Services and API Development

Course Code:	INF221	**Time Allowed:**	2 Hours 30 Minutes
Semester:	Semester Two – Year 2	**Total Marks:**	100 Marks

Student Name:	______________________________	**Student ID:**	______________

INSTRUCTIONS

1. This paper has THREE sections. Answer ALL questions in Sections A and B, and any TWO questions from Section C.
2. Write your answers in the spaces provided.
3. Marks for each question are indicated in brackets [ ].

SECTION A: SHORT ANSWER QUESTIONS [30 Marks]

Answer ALL questions in this section.

1. Define the term **Web Service**. [2]

2. State the full meaning of the following acronyms: [3]

(a) SOA

(b) REST

(c) HATEOAS

3. List **THREE** architectural styles used in web service design. [3]

4. State the **THREE** characteristics of each service in an SOA system. [3]

5. List the **FIVE** design requirements/principles of REST. [5]

6. Name the **FOUR** levels of the Richardson Maturity Model and state what each level introduces. [4]

7. State **THREE** data access problems associated with REST. [3]

8. List the **THREE** technologies/tools mentioned for interacting with web services. [3]

9. State the **THREE** layers of frontend web development and give one technology for each. [3]

SECTION B: DESCRIPTIVE QUESTIONS [40 Marks]

Answer ALL questions in this section.

10. Describe the concept of **Service Oriented Architecture (SOA)**. In your answer, explain its philosophy, the role of loose coupling and reusability, and how it supports integration across diverse systems. [8]

11. Describe the **six design constraints of REST**. For each constraint, provide a brief explanation of what it means in the context of a web service. [6]

12. Describe the **motivation** for web services. Your answer should cover the distributed nature of web technology, the need for interoperability, service composition and aggregation, and data consumption delivery mechanisms. [6]

13. Describe what **SOAP (Simple Object Access Protocol)** is and how it works. Include in your answer its 3-component model (Service Requester, Service Provider, Service Registry), its use of XML, and scenarios where it is best suited. [6]

14. Describe the difference between a **URI** and a **URL**. Then describe each part of the URL structure: scheme, host, path, query string, and anchor. [6]

15. Describe **gRPC** as an architectural style. In your answer, explain what Protocol Buffers are, how gRPC allows clients to invoke remote services as if they were local, and what the Protobuf IDL is used for. [8]

SECTION C: APPLICATION QUESTIONS [30 Marks]

Answer ANY TWO (2) questions from this section. Each question carries 15 marks.

Question 16

A university wants to build a student information system that will be accessed by mobile apps, web browsers, and third-party applications.

(a) Identify and justify which web service architectural style (REST, SOAP, or gRPC) you would recommend for this system. [5]

(b) Using the Richardson Maturity Model, describe how the university's API could evolve from Level 0 to Level 3. Give a concrete example at each level using a student resource (e.g., /students). [10]

Question 17

Consider the following URL:
https://api.university.ac.mw:8080/students/2024/results?year=2&sort;=DESC#grades

(a) Identify and label each component of the URL above (scheme, host, port, path, query string, anchor). [6]

(b) Explain the difference between the URI and URL of the resource above. [4]

(c) A developer complains that the endpoint returns far more data than the app needs. Name this problem and suggest how it could be resolved. [5]

Question 18

A fintech company is building a payment processing system that requires strict security, guaranteed message delivery, and integration with legacy banking systems.

(a) Explain why SOAP may be more appropriate than REST for this use case. [5]

(b) The company later decides to add a high-performance internal microservice for real-time fraud detection. Which architectural style would you recommend for this internal service and why? [5]

(c) Describe how the SOA principle of **service abstraction (blackbox)** protects the company's core business logic from being exposed to external consumers. [5]